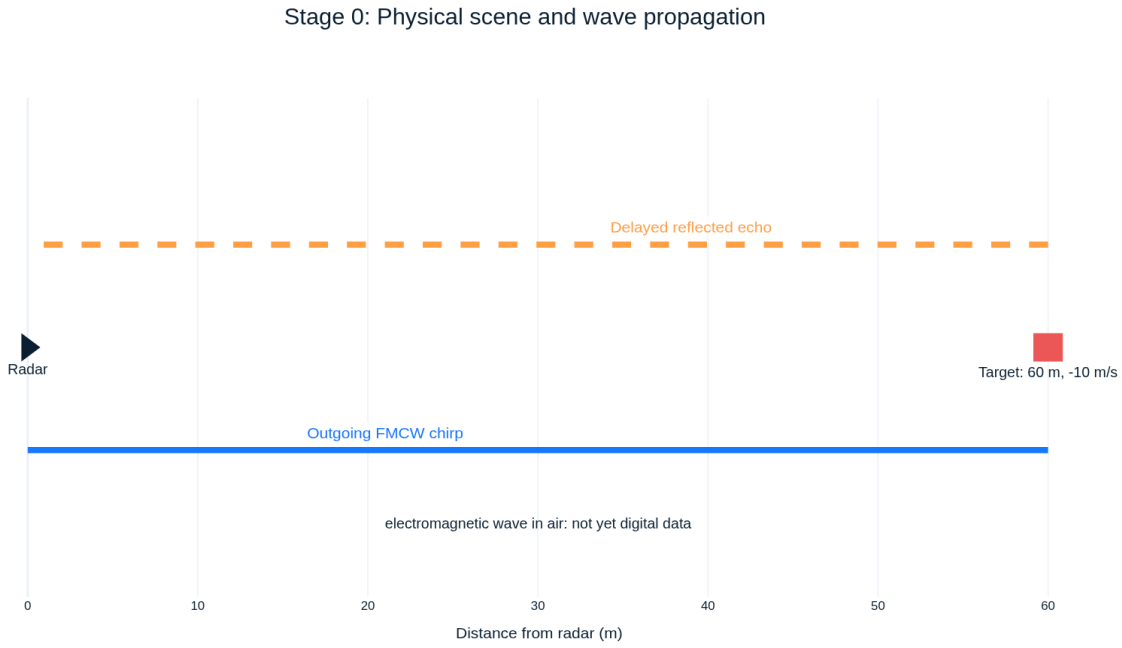
One Target Through the Entire FMCW Radar Pipeline

What the data looks like before and after every algorithm

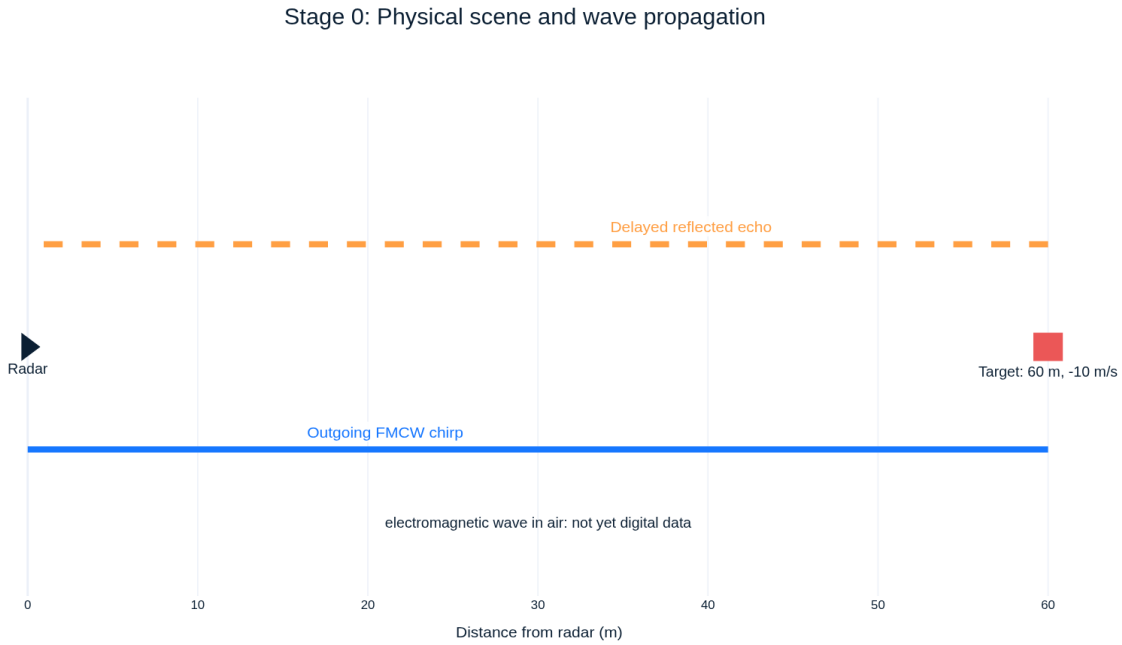


This version intentionally removes research-score comparisons. It follows one synthetic target at 60 m and -10 m/s through wave propagation, ADC, FFTs, detection, CDC, beamforming, AF detection, alignment, tracking, fusion, serialization, MF4 recording, decoding, and KPI comparison.

0. The order of the complete example



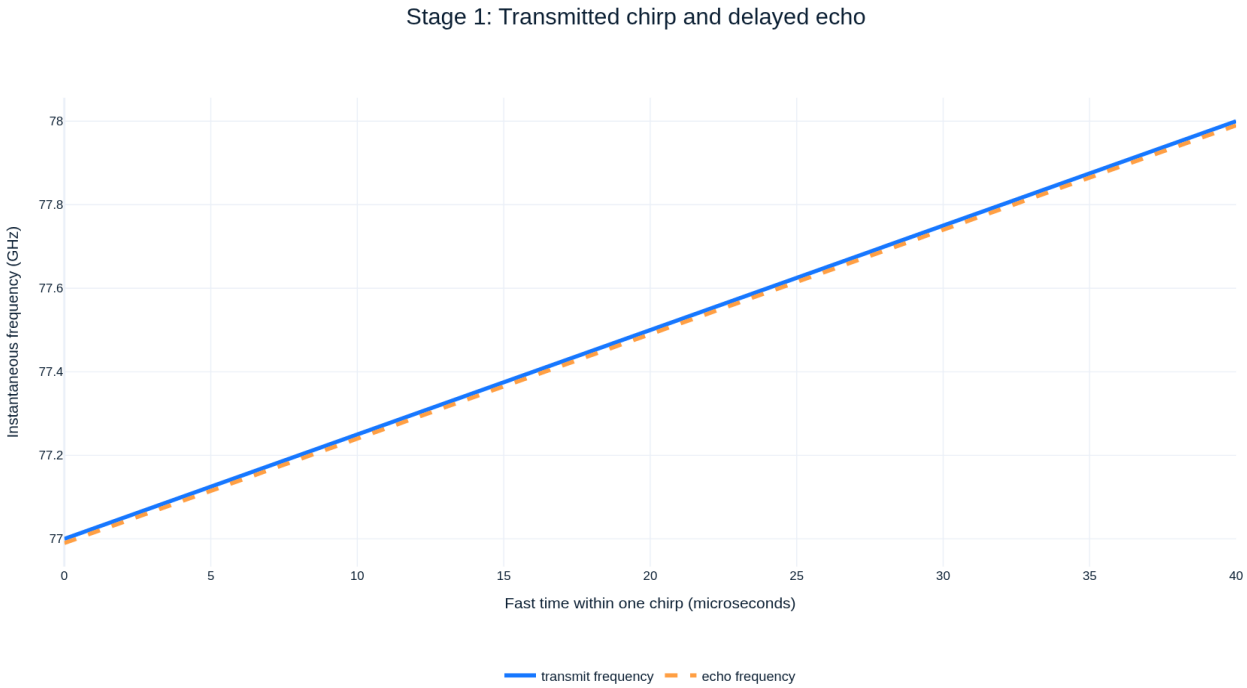
1. Before digital data: the electromagnetic wave



The transmitter launches a frequency-modulated wave. The target reflects only a small fraction. At this point there are no arrays, bins, objects, or files. There is only a field propagating through space, delayed and attenuated by the path.

What the algorithm does. The radar front end creates the chirp, amplifies it, receives the echo through antennas, and preserves enough phase information for later range, velocity, and angle estimation.

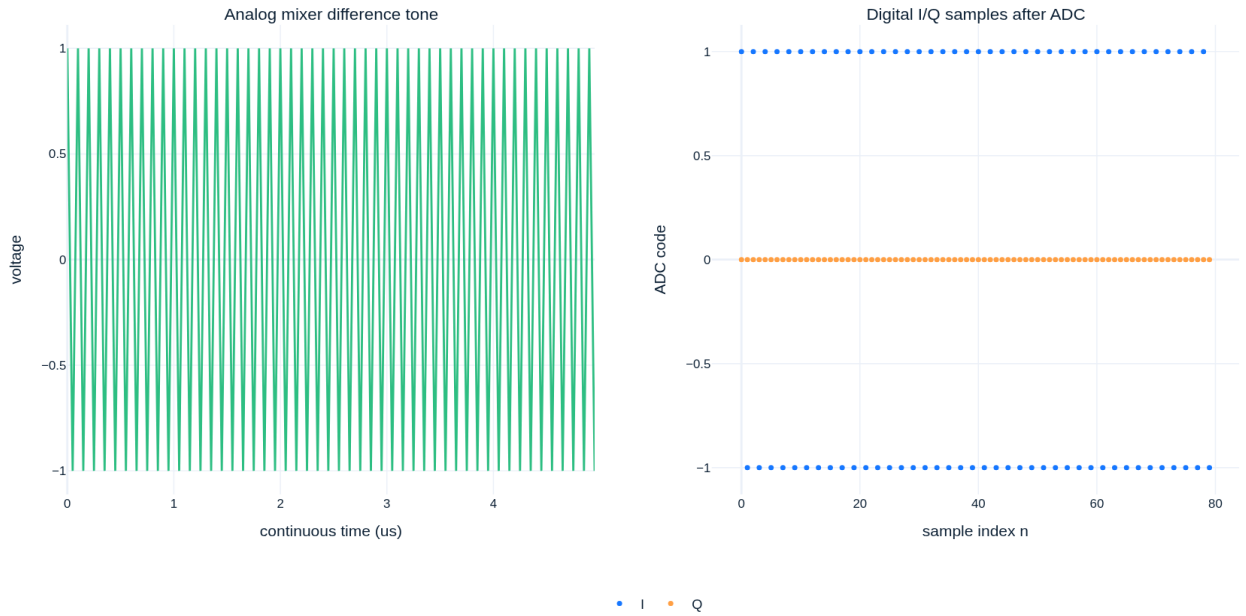
2. Chirp and echo: delay is encoded as frequency separation



The transmitted frequency rises linearly. The returned line has nearly the same slope but is delayed by tau=2r/c. For the 60 m example, the beat approximation is f_b=S tau=10 MHz.

What the algorithm does. The mixer compares the current transmit phase with the older received phase. It converts a 77 GHz RF problem into a megahertz beat that an ADC can sample.

3. Mixer output and I/Q ADC

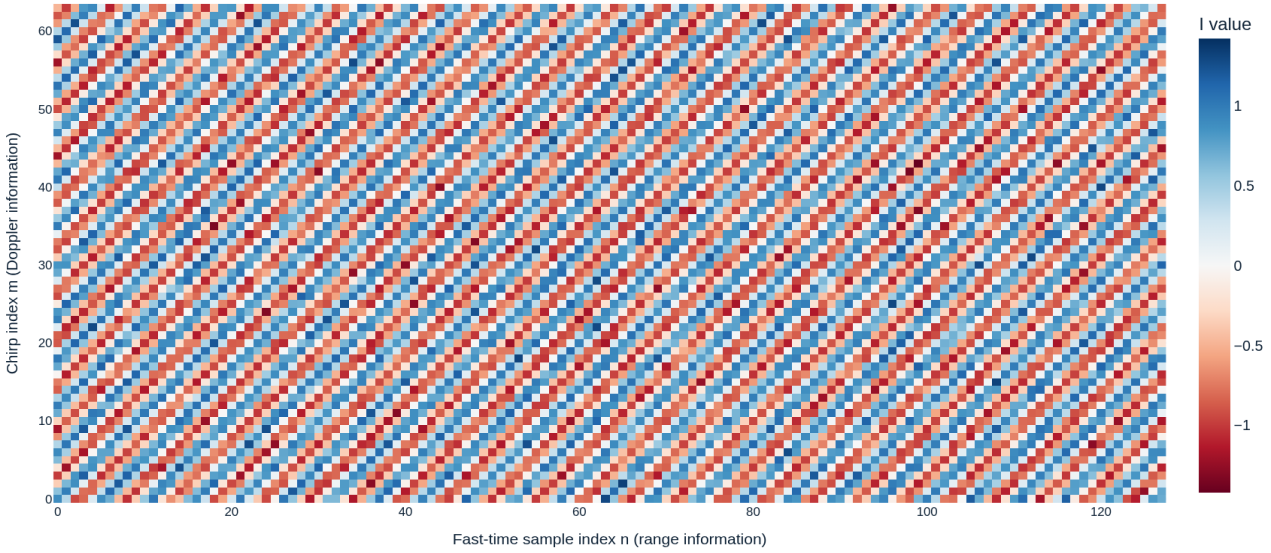


Before the ADC, the low-frequency beat is an analog voltage. After the ADC, one chirp is a sequence of I and Q integer samples. The pair is interpreted as x[n]=I[n]+jQ[n].

What the algorithm does. The low-pass filter rejects the mixer sum term. Quadrature sampling keeps complex phase and the sign of rotation, which is required for signed frequency and coherent processing.

4. Raw frame before any FFT

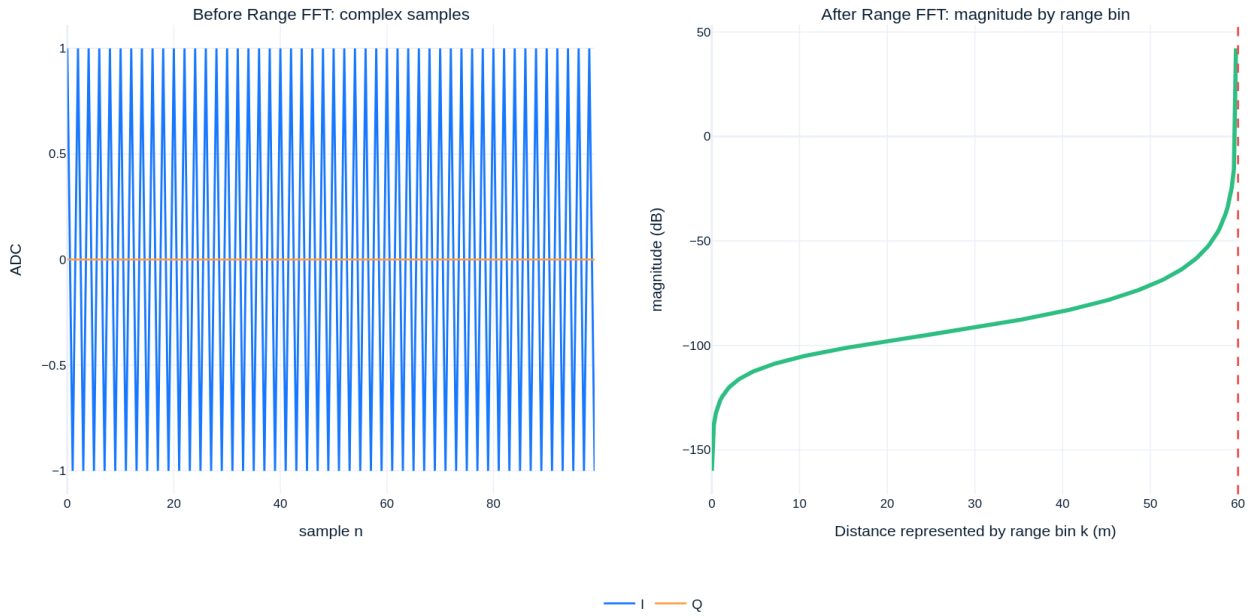
Stage 2: Raw ADC frame before FFT processing



The frame is now a matrix. Columns are fast-time samples inside a chirp. Rows are repeated chirps. Range and velocity are present, but they are hidden as oscillation and phase progression rather than explicit metres or metres per second.

What the algorithm does. No detector should operate on this picture as if it were a Range-Doppler map. The next two transforms apply different spectral analysis along the two axes.

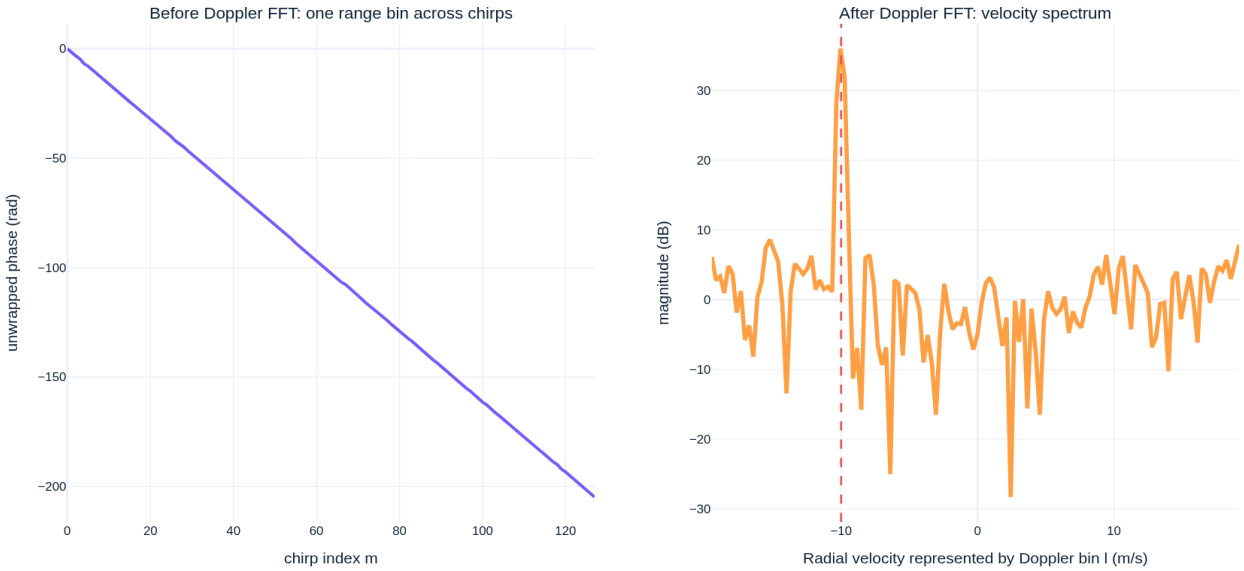
5. Range FFT: samples become range bins



The left side is still a waveform. The right side contains spectral peaks indexed by range bin k and mapped to distance using the chirp slope, sample interval, FFT size, and speed of light.

What the algorithm does. For every chirp and receive channel: apply a window, compute the FFT across fast time, retain complex output, calibrate bin-to-range scaling, and optionally refine the peak between bins.

6. Doppler FFT: chirp-to-chirp phase becomes velocity

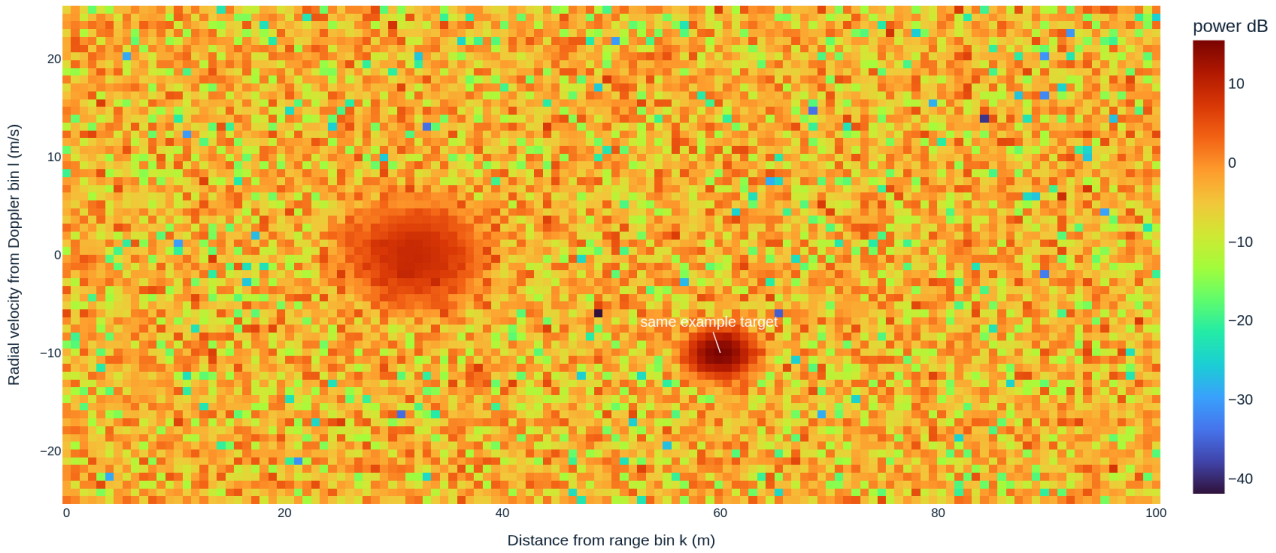


A chosen range bin is followed across all chirps. Constant radial velocity produces nearly constant phase advance. The Doppler FFT changes that phase ramp into a peak indexed by l and mapped to radial velocity.

What the algorithm does. For every range bin and channel: window across slow time, compute the FFT over chirps, shift or reinterpret signed bins, and attach the velocity scale. Fast-time and slow-time axes must never be swapped.

7. After both FFTs: the Range-Doppler matrix

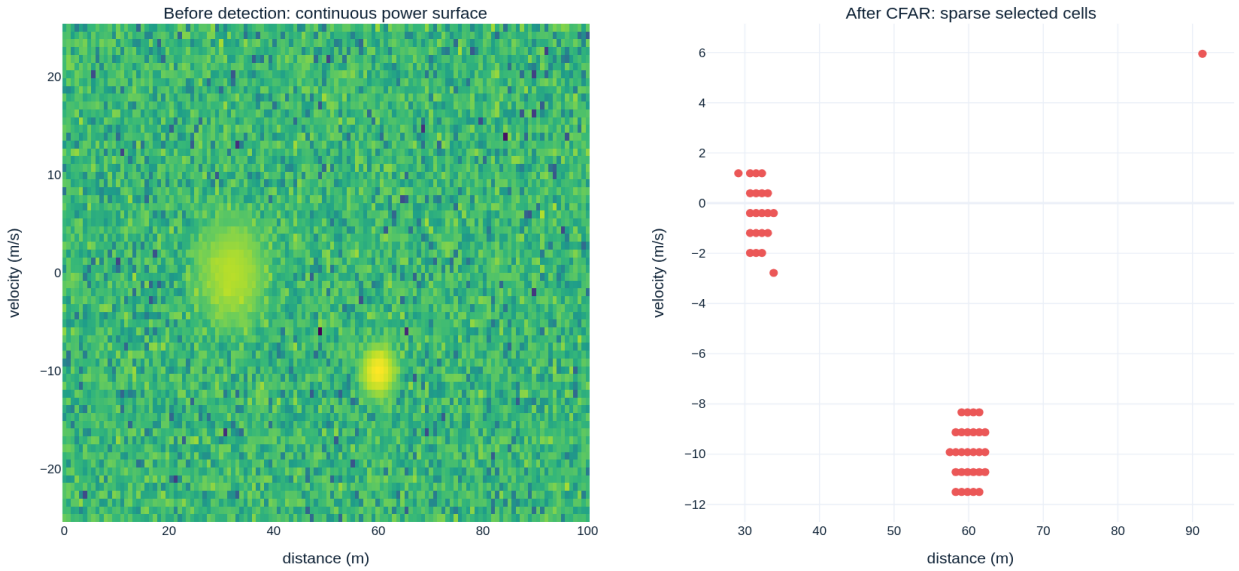
Stage 4: Two-dimensional Range-Doppler map



Now each cell has two physical coordinates: distance and radial velocity. The example target appears near 60 m and -10 m/s. A second static structure is shown near zero Doppler. Angle is still not known.

What the algorithm does. Combine or integrate channel energy to form a detection surface while retaining channel-level complex values for later angle estimation. Apply calibration and normalization consistently.

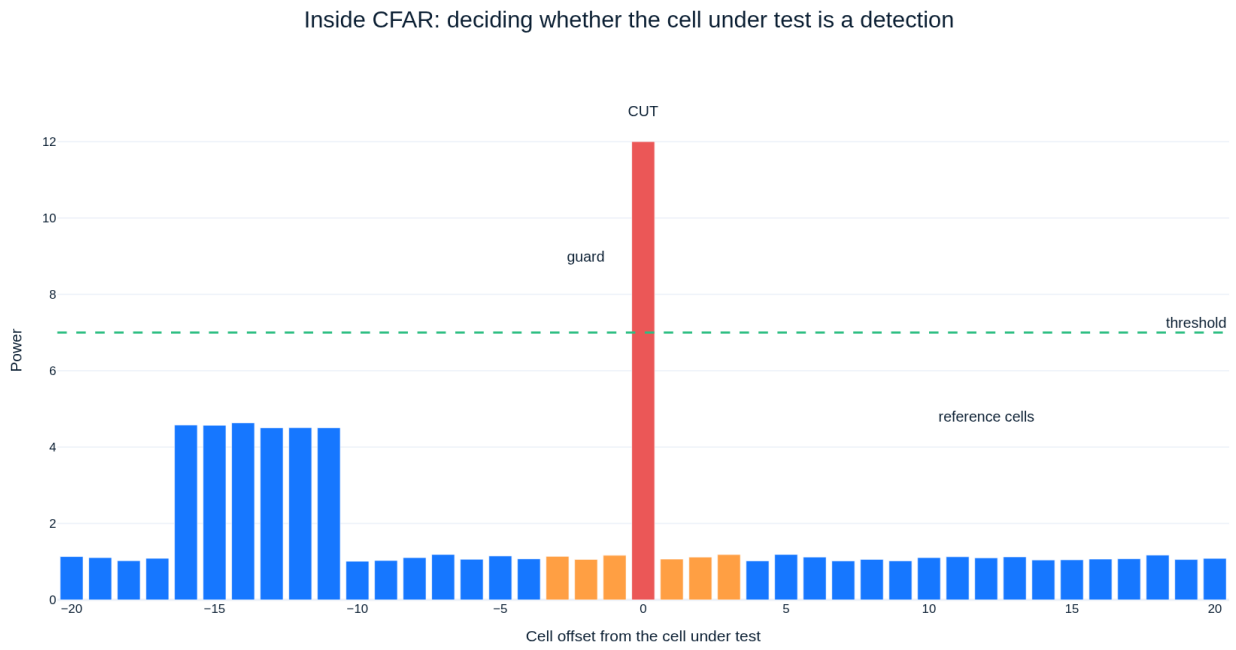
8. Detection: dense power becomes sparse cells



Before CFAR, every grid location has a power value. After CFAR, only cells judged unusually strong relative to their local background are kept as candidates.

What the algorithm does. Estimate background from reference cells, protect the cell under test with guard cells, calculate the threshold for the selected false-alarm model, compare the CUT, and optionally group adjacent detections.

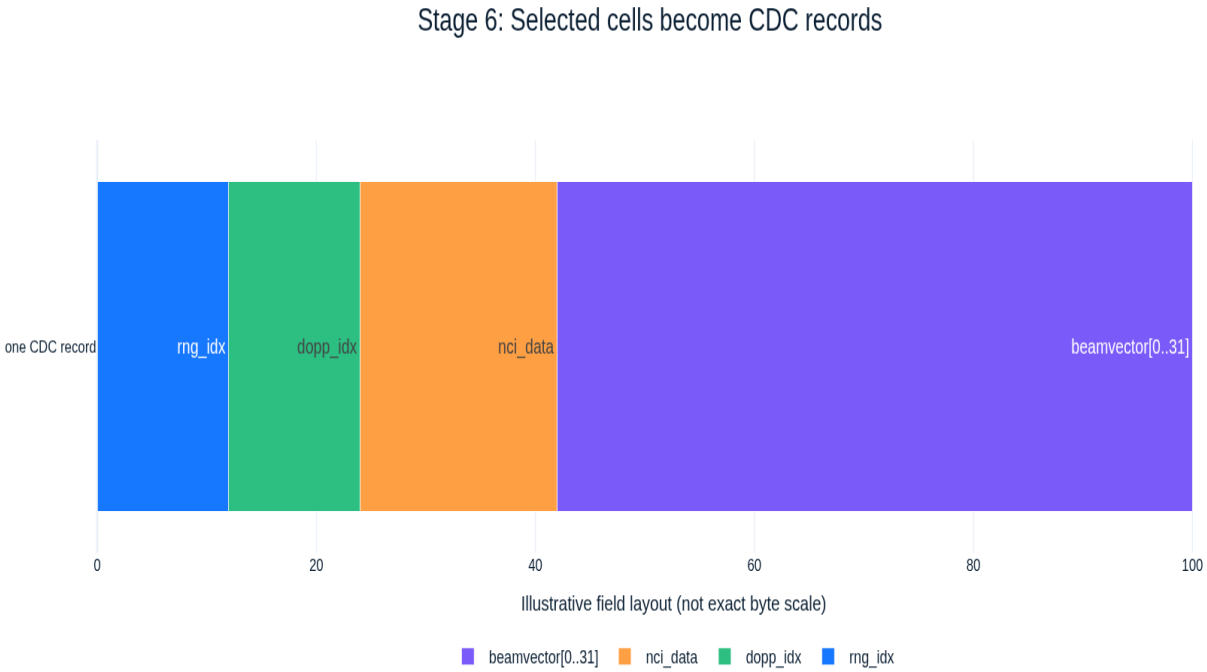
9. Inside one CFAR decision



The red CUT is compared with a green threshold. Orange guard cells are ignored so the target does not raise its own background estimate. Blue cells estimate the surrounding clutter or noise.

What the algorithm does. The algorithm must account for the effective number of reference cells, nonhomogeneous clutter, censored cells, window edges, and the fact that one physical target may occupy several bins.

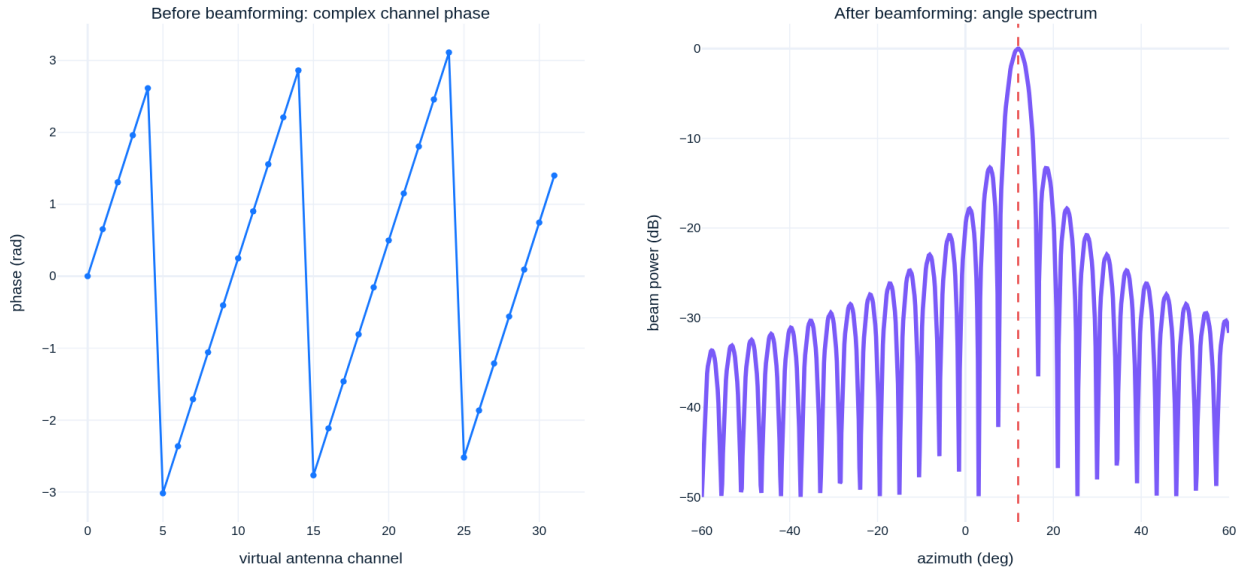
10. CDC: sparse detections become transport records



The dense tensor is too large to send downstream. Selected cells become records containing range index, Doppler index, energy, and a multi-channel beamvector. The list can be stored or split across UDP datagrams.

What the algorithm does. Choose which cells survive, preserve their channel information, order or prioritize records, enforce capacity, and attach frame metadata. Overflow policy is a data-quality decision, not merely a memory detail.

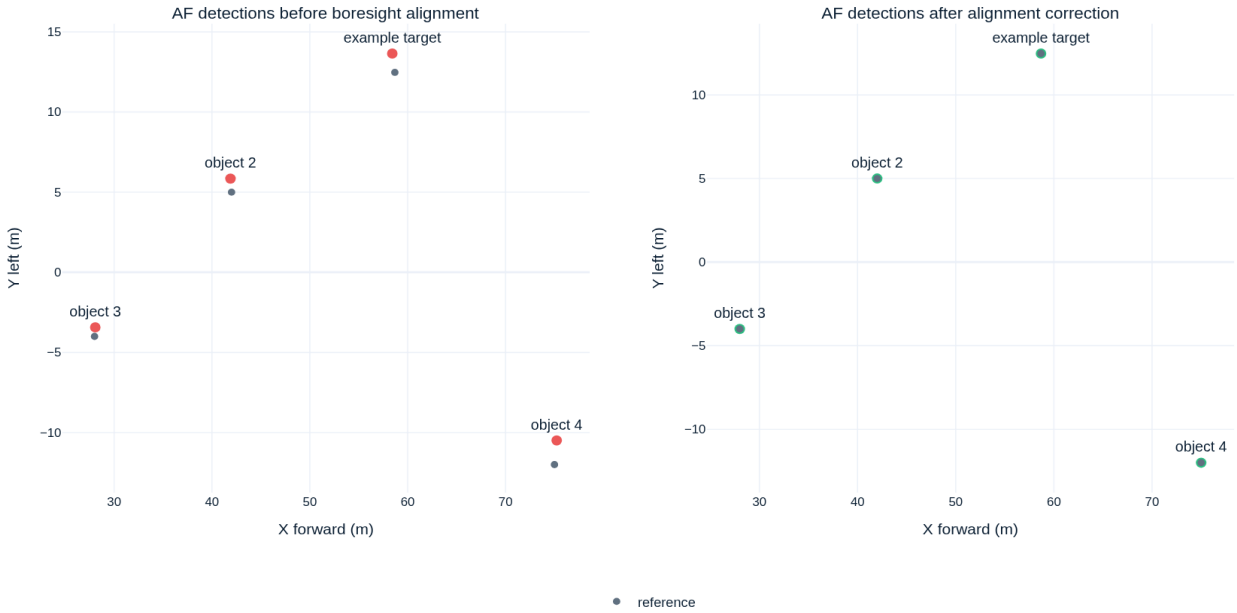
11. Beamforming: channel phase becomes angle



The same detected Range-Doppler cell has a complex value for every virtual antenna. A plane wave produces an ordered phase progression across channels. Beam steering converts that progression into an azimuth or elevation spectrum.

What the algorithm does. Calibrate channel phase and amplitude, evaluate steering vectors or an angle FFT, find angular peaks, refine them, and convert the selected angle into a physical detection field.

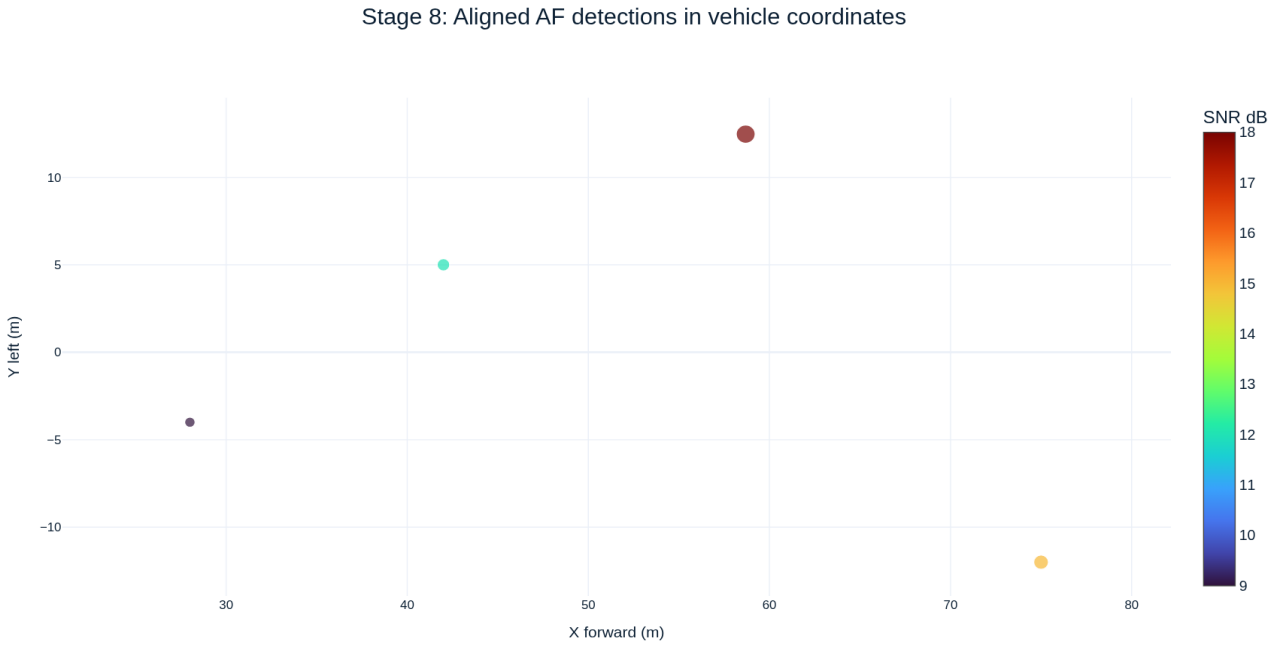
12. AF detection first, then AF alignment correction



The AF detection already contains range, radial velocity, azimuth, elevation, SNR, and related fields. However, mounting or dynamic boresight bias can rotate the detections. Alignment estimates or reads the correction and transforms detections into the intended sensor or vehicle orientation.

What the algorithm does. Static channel calibration happens before or during angle estimation. Dynamic AF alignment is a later geometric correction using calibration state, suitable references, and quality gates. After correction, detections should align with the vehicle frame and other sensors.

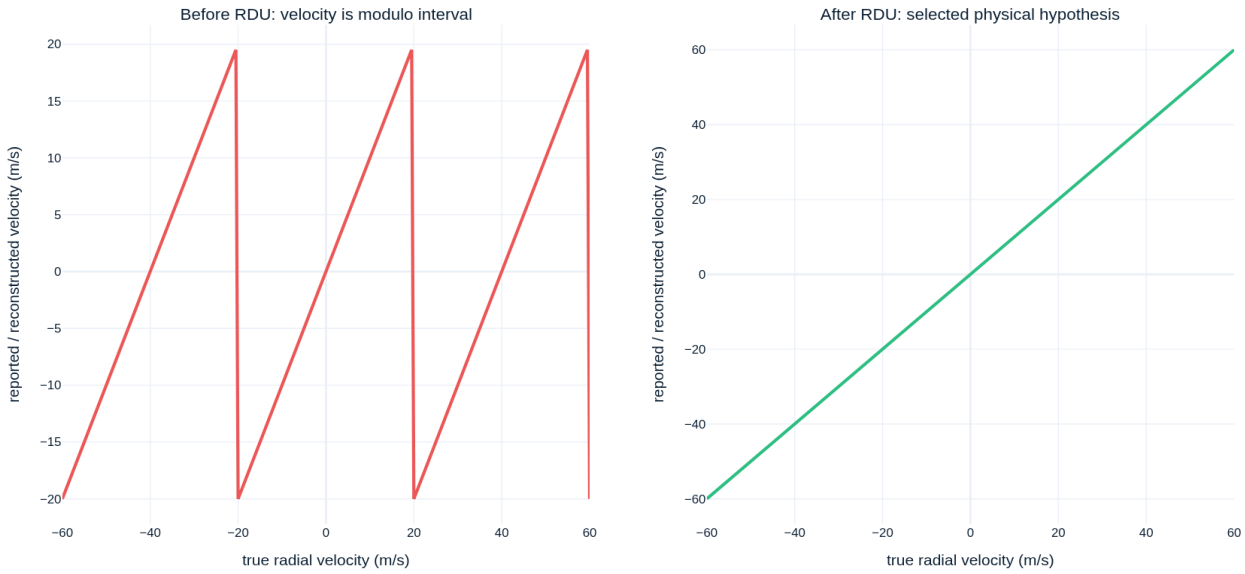
13. What an aligned AF detection list looks like



The output is no longer a spectral image. It is a variable-length list of physical observations. Each point can be plotted in Cartesian space, but the stored record should retain timestamp, frame, range, velocity, angles, SNR, flags, and uncertainty-relevant metadata.

What the algorithm does. Convert polar fields to the required coordinate frame, preserve original values when needed for debugging, reject invalid calibration states, and maintain a clear schema version.

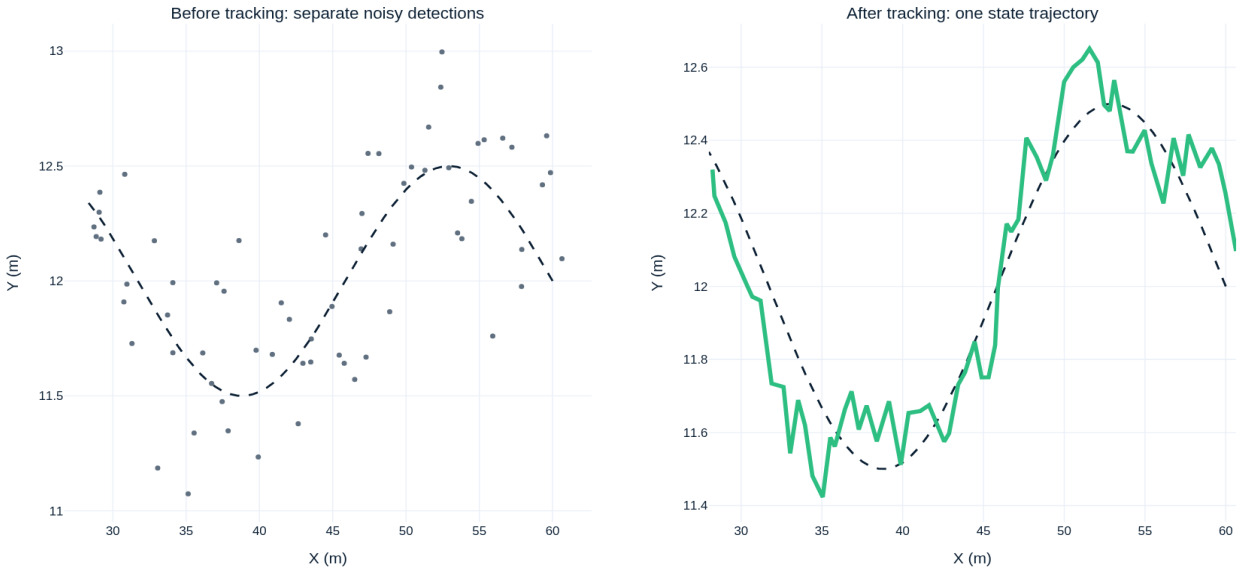
14. RDU and post-processing



The Doppler FFT measures velocity modulo an unambiguous interval. A very fast target can wrap to the wrong sign or magnitude. RDU selects the physical hypothesis using staggered sequences, consistency checks, and scene or track information.

What the algorithm does. Generate candidate unwrapped velocities, score consistency across sequences, choose only when identifiable, and mark ambiguity rather than silently inventing a velocity.

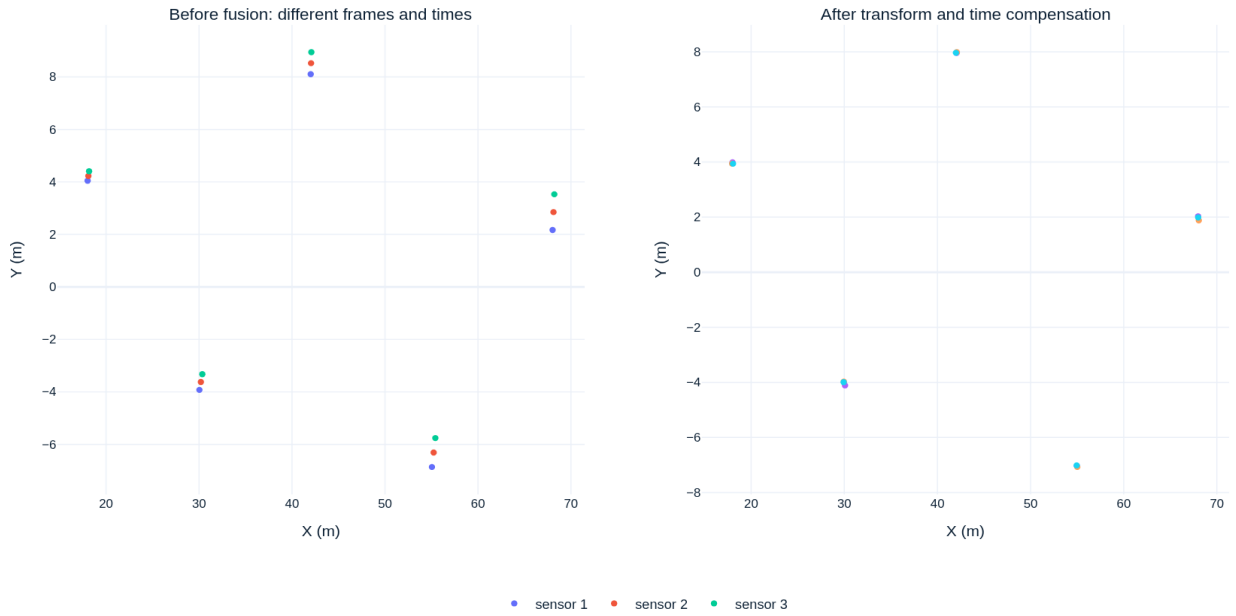
15. Tracking: detections become objects over time



Before tracking, each frame contains separate noisy points. After tracking, measurements are associated through time into an object state with position, velocity, covariance, age, status, and ID.

What the algorithm does. Predict with the motion model, form measurement innovations, gate candidates, associate detections, update with EKF or another estimator, manage birth and death, and propagate covariance with numerically stable equations.

16. Multi-radar compensation and fusion

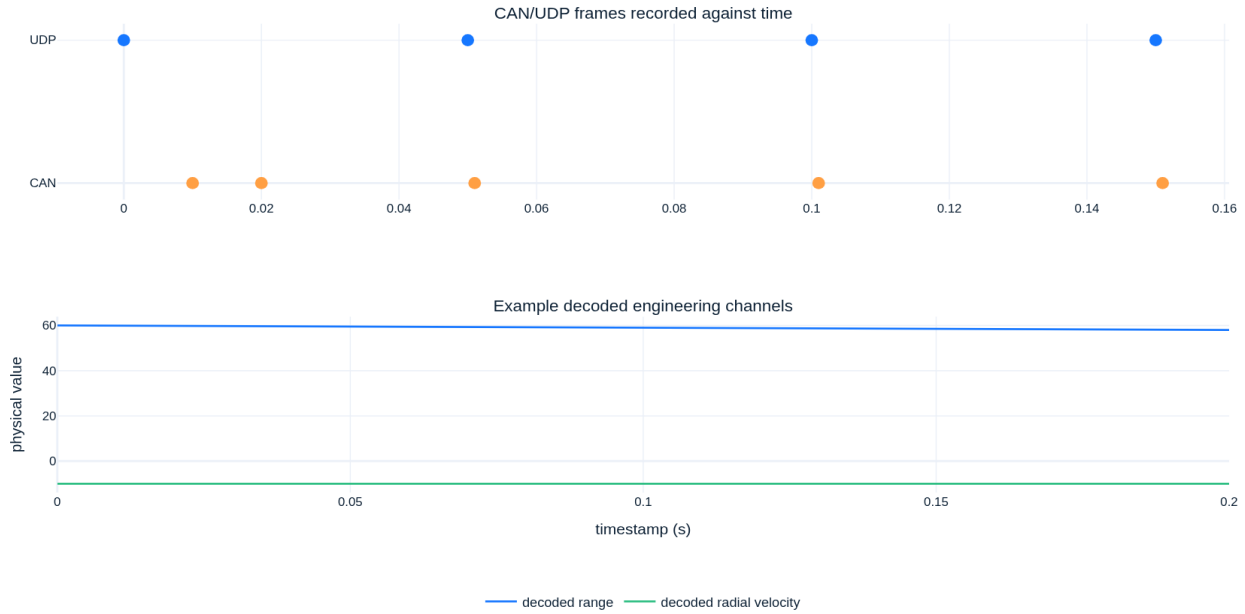


Different radars measure the world from different locations and at slightly different times. Directly overlaying their points smears one physical object. After extrinsic transformation and timestamp compensation, corresponding observations occupy a common frame and epoch.

What the algorithm does. Transform sensor coordinates into VCS, deskew with ego translation and rotation, propagate uncertainty, associate cross-sensor observations, and fuse them without double-counting correlated information.

17. Serialization and MF4: where recorded data appears

Stage 12: Conceptual MF4 recording timeline and decoded channels

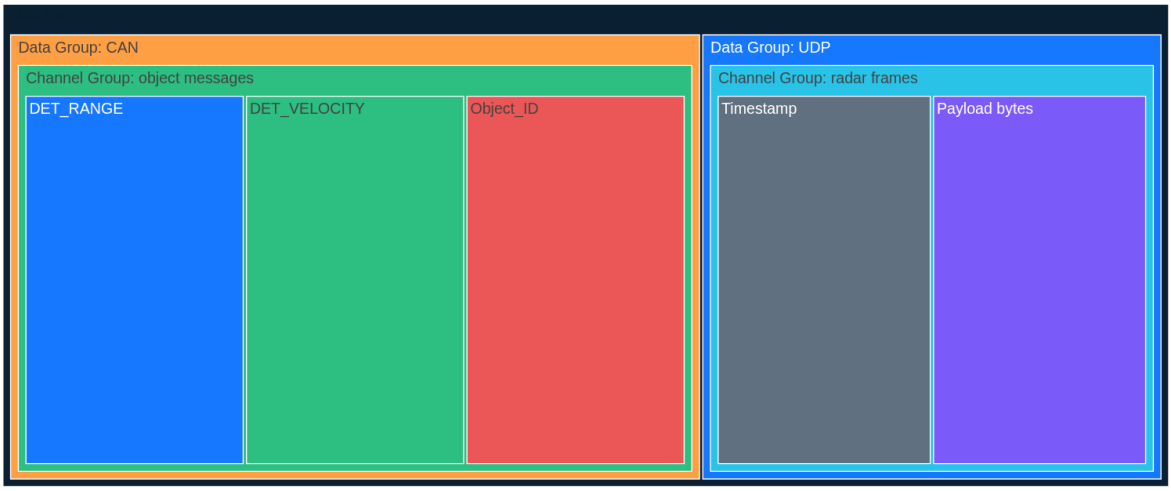


An MF4 file is a time-indexed measurement container. In a re-simulation recording it commonly stores CAN and Ethernet/UDP traffic, timestamps, channel groups, and decoded engineering channels when available. It does not automatically mean raw ADC samples are present. Raw signal-processing vectors may instead be stored in separate binary files or specialized groups.

What recording does. It preserves time order, bus metadata, message IDs, payload bytes, channel definitions, scaling information, and synchronization. Decoders later turn these transport records into radar-specific structures.

18. Conceptual MF4 contents

Conceptual MF4 hierarchy: groups, timestamps, payloads, and channels



Think of MF4 as a container with data groups and channel groups. One group may contain raw UDP frames from a radar stream, another CAN object messages, and others decoded channels. Each sample is tied to a master timestamp. The exact layout depends on the logger configuration and project schema.

Important distinction. “Stored in MF4” can mean raw payload bytes, decoded signal channels, or both. Before designing an algorithm, inspect the channel groups and determine whether the needed level is ADC, CDC, AF detection, track output, or CAN object output.

19. After decoding: HDF5 or CSV view

After decoding: algorithm-friendly HDF5 hierarchy



A decoder changes timestamped bus frames into algorithm-friendly arrays. HDF5 is useful for nested sensor groups, frame arrays, CDC records, AF detections, and candidate outputs. CSV is useful for flat KPI tables but is less suitable for variable-length vectors and complex arrays.

Decoder responsibilities. Parse packets, validate length and counters, apply schema versions, reconstruct frames, attach timestamps, scale integer fields, preserve invalid flags, and write a deterministic hierarchy. The decoder must not silently perform the candidate algorithm.

20. Example schemas at each stored level

Level	Example stored fields	Best visualization
Raw bus in MF4	time, bus/channel, frame ID, payload bytes	timeline and byte inspector
Decoded CDC in HDF5	rng_idx, dopp_idx, nci_data, beamvector	Range-Doppler scatter plus channel phase
AF detection	time, r, vr, az, el, SNR, flags	Range-Azimuth or XY point cloud
Alignment state	time, azimuth bias, elevation bias, validity	bias versus time, before/after overlay
Track output	ID, x, y, vx, vy, covariance, status	trajectory plus covariance ellipse
CAN object output	message ID, raw fields, decoded values	timeline and object table
KPI pair table	reference ID, candidate ID, errors, latency	error distributions and pass/fail

The data level determines the algorithm you can test. CFAR requires a power map or equivalent CDC context. Beamforming requires per-channel complex values. Alignment requires AF detections plus calibration state. Tracking requires time-ordered detections. KPI-only CSV cannot reconstruct missing upstream phases.

21. End-to-end algorithm contract

Algorithm	Input representation	Output representation	Main operation
Mixer/ADC	analog RF	I/Q sample arrays	down-convert and sample
Range FFT	I/Q by sample	complex range bins	FFT over fast time
Doppler FFT	range bins by chirp	Range-Doppler bins	FFT over slow time
CFAR	power grid	candidate cell list	local statistical threshold
CDC pack	selected cells and channel values	transport records	compress, prioritize, serialize
Beamforming	beamvector per selected cell	angle peaks	coherent steering / angle FFT
AF alignment	AF detections plus bias state	corrected AF detections	rotate and validate
RDU	wrapped Doppler observations	velocity hypotheses	unwrap and disambiguate
Tracking	detections over time	states, IDs, covariance	predict, associate, update
Fusion	multi-sensor states at different times	common-epoch state	transform, deskew, combine
Serialization	physical fields	CAN/UDP bytes	scale, quantize, pack
MF4 logger	timestamped frames/channels	measurement file	index and store
Decoder	MF4 bus data	HDF5/CSV arrays	parse and reconstruct
KPI	reference and candidate outputs	scores and error tables	pair and compare

22. The central intuition

Every algorithm changes the representation. The easiest way to understand the radar pipeline is not by memorizing function names, but by always asking: what are the axes before the function, what mathematical structure is hidden in them, what new axes or fields exist after the function, and what information was discarded?

Before	Hidden information	Algorithm	After
Wave in air	delay, Doppler, arrival angle	RF front end and mixer	analog beat
Analog beat	phase and frequency	I/Q ADC	complex samples
Samples inside chirp	beat frequency	Range FFT	range bins
Same range bin across chirps	phase progression	Doppler FFT	velocity bins
Dense spectral grid	statistical peaks	CFAR	sparse cells
Per-channel complex phases	arrival direction	Beamforming	angles
Biased physical detections	mounting orientation	AF alignment	aligned detections
Noisy frames	temporal continuity	Tracker	object state
Different sensors/times	shared world geometry	Fusion	common state
Physical values	wire contract	Serialization/MF4	timestamped bytes and channels

This is the version to use for learning and quizzes. Research improvements, novelty scores, recall comparisons, and patent discussions are deliberately excluded so they do not interrupt the understanding of the base data flow.

23. Source and scope note

The processing order, equations, data types, MF4 replay context, CDC records, AF detections, alignment, tracking, fusion, CAN serialization, and KPI concepts are based on the supplied radar monograph and experiment ledger. All plotted signals and file contents in this guide are synthetic examples designed to explain representation changes. They are not claimed to be a dump of an uploaded real MF4 file.

To create a fully repository-specific edition, an actual MF4 or HDF5 sample would be inspected and the conceptual schemas on pages 17-20 would be replaced with its real channel groups, dataset names, array shapes, timestamps, and example records.